

Republic of the Philippines
PROFESSIONAL REGULATION COMMISSION
Manila

Board of CIVIL ENGINEERING

CIVIL ENGINEERING Licensure Examination

Monday, September 21, 2026

8:00 AM – 2:00 PM

PRINCIPLES OF STRUCTURAL ANALYSIS AND DESIGN

SET A

INSTRUCTION: Select the correct answer for each of the following questions. Mark **only one answer** for each item by shading the box corresponding to the letter of your choice on the answer sheet provided. **STRICTLY NO ERASURES ALLOWED.**

MULTIPLE CHOICE

SITUATION 1 – A standard steel pipe is used as a pole. It is fixed on the ground and is subjected to the lateral force.

Dimensions:

Outside diameter of the pipe, $d_o = 324 \text{ mm}$

Inside diameter of the pipe, $d_i = 304 \text{ mm}$

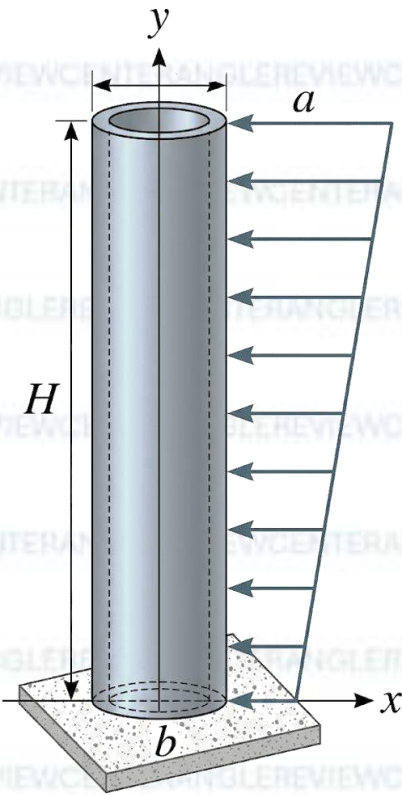
Height, $H = 12 \text{ m}$

Lateral force:

$a = 2.88 \text{ kN/m}$

$b = 1.44 \text{ kN/m}$

Weight of the pipe, $W = 0.28 \text{ kN/m}$



- Which of the following gives the MAXIMUM shear force (kN)?
A. 25.92
B. 17.28
C. 34.56
D. 8.64
- Which of the following gives the resulting moment at the base of the pole (kN-m)?
A. 172.80
B. 155.52
C. 138.24
D. 103.68
- Which of the following gives the resulting compressive stress (MPa) in the pole due to all loads?
A. 230.36
B. 230.02
C. 207.36
D. 229.68

SITUATION 2 – A torque wrench is used to tighten the pipe shown.

Dimensions:

$S_1 = 400 \text{ mm}$

$S_2 = 250 \text{ mm}$

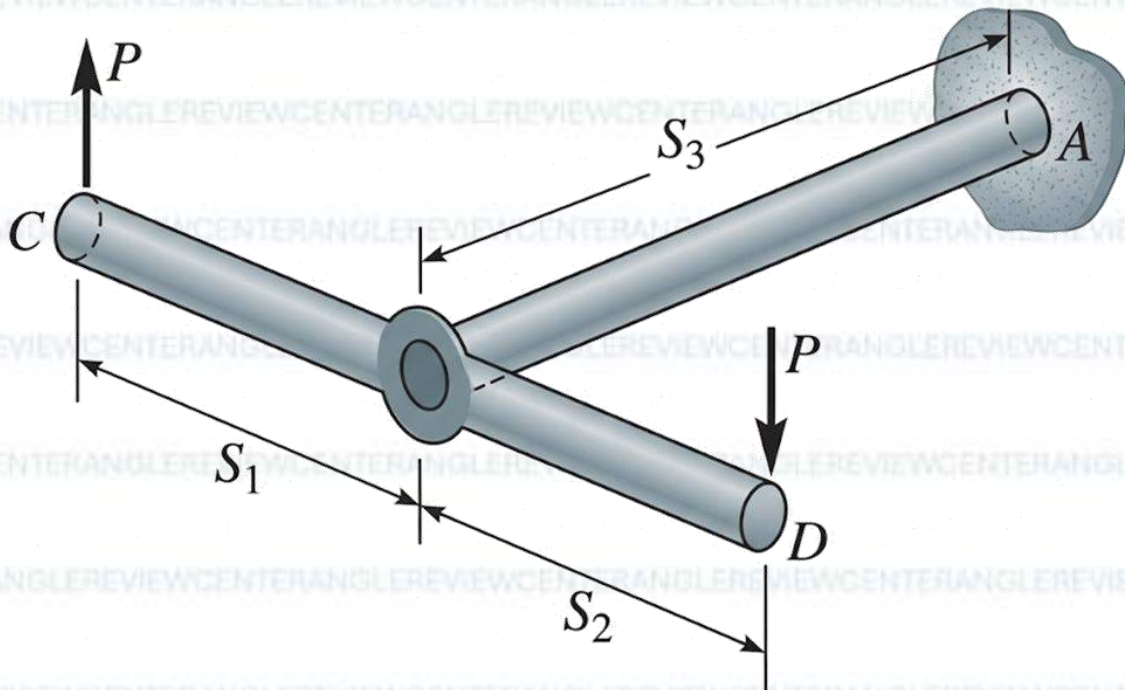
$S_3 = 100 \text{ mm}$

Allowable shear stress = 60 MPa

Modulus of rigidity, $G = 78 \text{ MPa}$

Moment of inertia = $(\pi)^4 / 32$

ANGLE of twist = $T \cdot L / J \cdot G$



4. Given:

Diameter of solid pipe = 20 mm

How much is the MAXIMUM force P (N) that can be applied based on the allowable shear stress?

A. 236

B. 145

C. 628

D. 73

5. Given:

Outside diameter of hollow pipe = 50 mm

Thickness of hollow pipe = 6 mm

How much is the MAXIMUM force P (kN) that can be applied such that the angle of twist at A does not exceed 5 degrees.

A. 25.7

B. 42.8

C. 2453.0

D. 69.6

6. Given:

The torque applied to tighten the hollow pipe = 200 N-mm

Outside diameter of hollow pipe = 50 mm

Thickness of hollow pipe = 6 mm

How much is the resulting MAXIMUM shear stress (MPa) in the pipe.

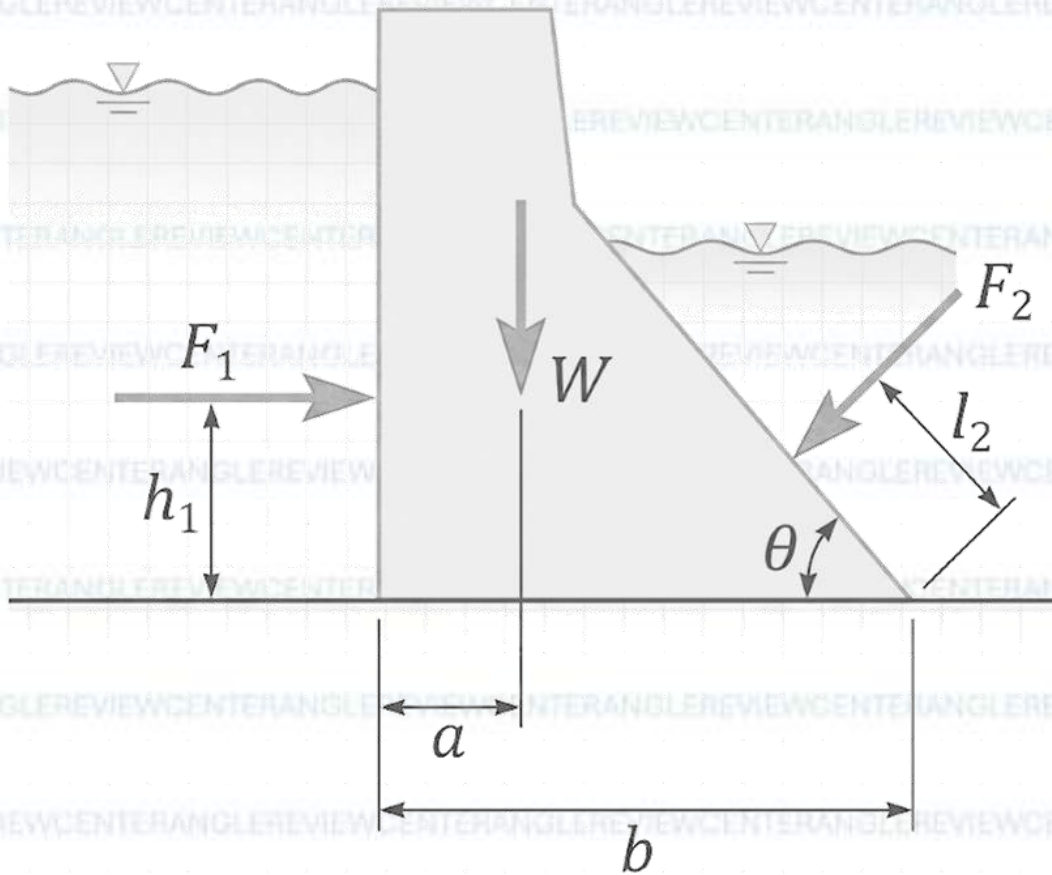
A. 9.3

B. 12.2

C. 8.2

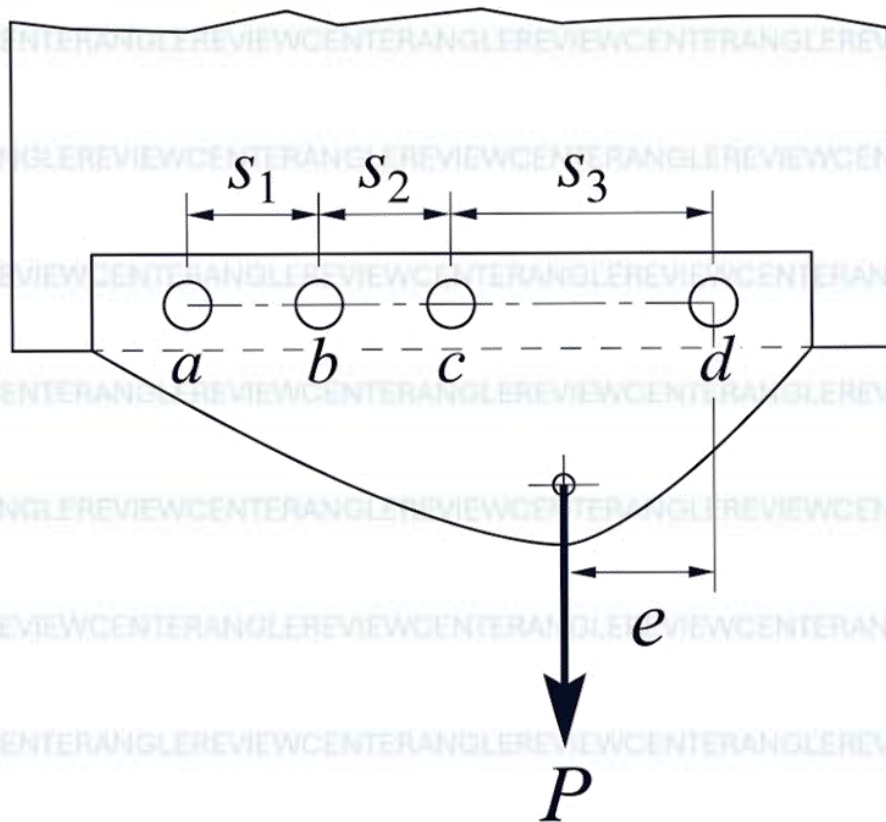
D. 20.4

This review material is solely for ANGLE reviewees. Any unauthorized use, copying, distribution, sharing, or anything related to cybercrime is strictly forbidden and will be subject to legal action.

$L_2 = 1.8 \text{ m}$ 

7. Given:
Coefficient of friction at the base of the dam = 0.3
Solve for the factor of safety against sliding.
- A. 1.66
B. 1.49
C. 0.80
D. 1.14
8. Determine the eccentricity (m) of the resultant force at the base of the dam.
- A. 0.27
B. 0.45
C. 0.12
D. 0.68
9. What is the height (m) of water at the tailwater (above the toe) of the dam?
- A. 5.40
B. 4.68
C. 3.60
D. 7.49

Load, $P = 55 \text{ kN}$



10. Determine the shear stress (MPa) in bolt “d”.
 A. 31.8
 B. 107.4
 C. 101.3
 D. 75.6
11. Determine the shear stress (MPa) in bolt “b”.
 A. 22.5
 B. 54.4
 C. 11.9
 D. 33.2
12. If the line of action of load P coincides with the center of gravity of the bolts, find the MAXIMUM shear stress (MPa) in the bolts.
 A. 75.6
 B. 175.1
 C. 10.9
 D. 43.8

SITUATION 5 – A wide flange section is used as a purlin between roof trusses 6 m on centers. The purlins have a spacing of 1.20 m on centers.

Given:

Wide flange properties:

Area, $A = 10,900 \text{ mm}^2$

Depth, $d = 310 \text{ mm}$

Flange width, $bf = 250 \text{ mm}$

Flange thickness, $tf = 16 \text{ mm}$

Thickness of web, $tw = 9 \text{ mm}$

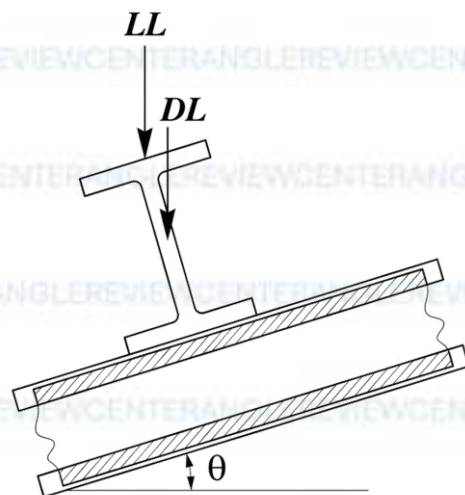
$I_x = 198 \times 10^6 \text{ mm}^4$

$I_y = 44 \times 10^6 \text{ mm}^4$

Dead load = 1.44 kPa (acting on centroid)

Live load = 1.0 kPa (acting on flange)

angle of inclination = 26.6 degrees



This review material is solely for ANGLE reviewees. Any unauthorized use, copying, distribution, sharing, or anything related to cybercrime is strictly forbidden and will be subject to legal action.

13. Calculate the flexural stress (MPa) due to bending about the x-axis.
 A. 9.23
 B. 10.23
 C. 7.68
 D. 18.46
14. Calculate the flexural stress (MPa) due to bending about the y-axis.
 A. 23.65
 B. 16.74
 C. 33.49
 D. 17.22
15. Calculate the MAXIMUM horizontal shear stress (MPa) about the x-axis.
 A. 2.98
 B. 2.82
 C. 0.11
 D. 2.59

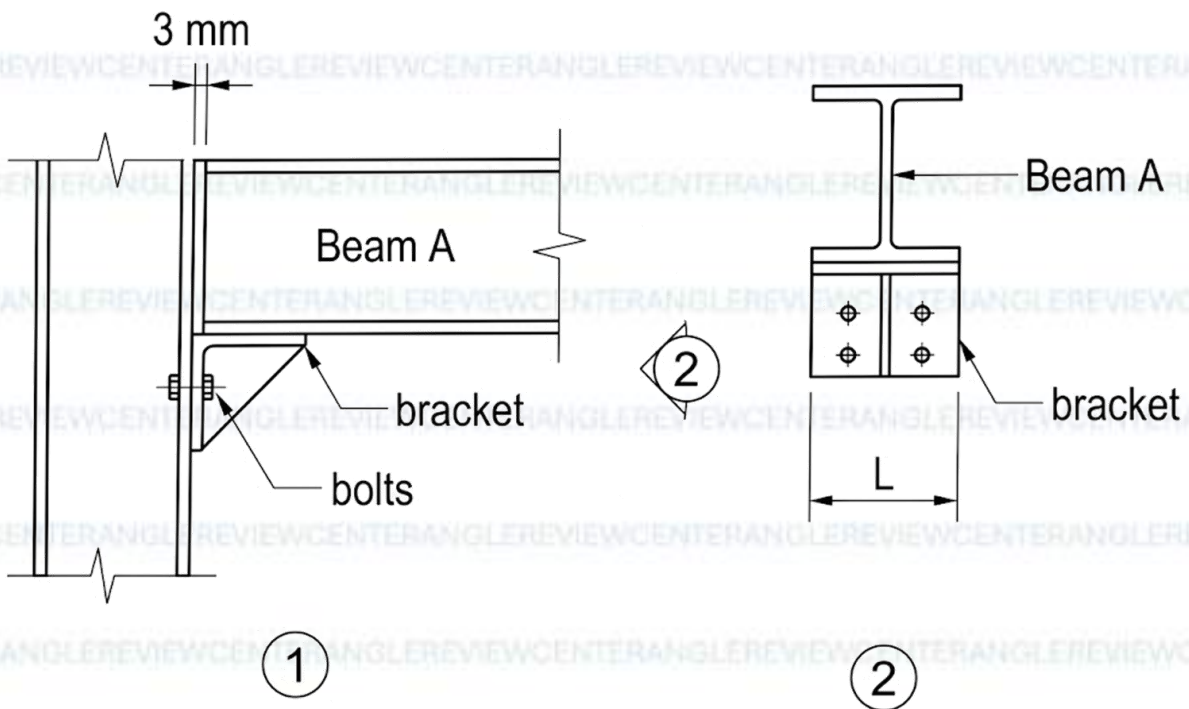
SITUATION 6 – The ANGLE bracket supports beam A. The bracket is bolted to the flange of the supporting column with 2 – 20 mm diameter bolts. The load from Beam A = 60 kN. Assume that the load is concentrically applied at the connection.

Given:

Bracket = 75 mm x 75 mm x 12 mm thick, L = 155 mm

Thickness of column flange = 125 mm

Thickness of beam flange = 10 mm



16. Find the CRITICAL bolt bearing stress (MPa).
 A. 150
 B. 100
 C. 125
 D. 250
17. Find the CRITICAL bolt shear stress (MPa).
 A. 169.8
 B. 382.0
 C. 265.2
 D. 95.5
18. Find the bearing pressure (MPa) in the bracket.
 A. 2.6
 B. 5.4
 C. 1.8
 D. 4.0

This review material is solely for ANGLE reviewees. Any unauthorized use, copying, distribution, sharing, or anything related to cybercrime is strictly forbidden and will be subject to legal action.

SITUATION 7 – The flagpole is braced by three cables as shown. The resultant of the tensile forces acts along the vertical axis.

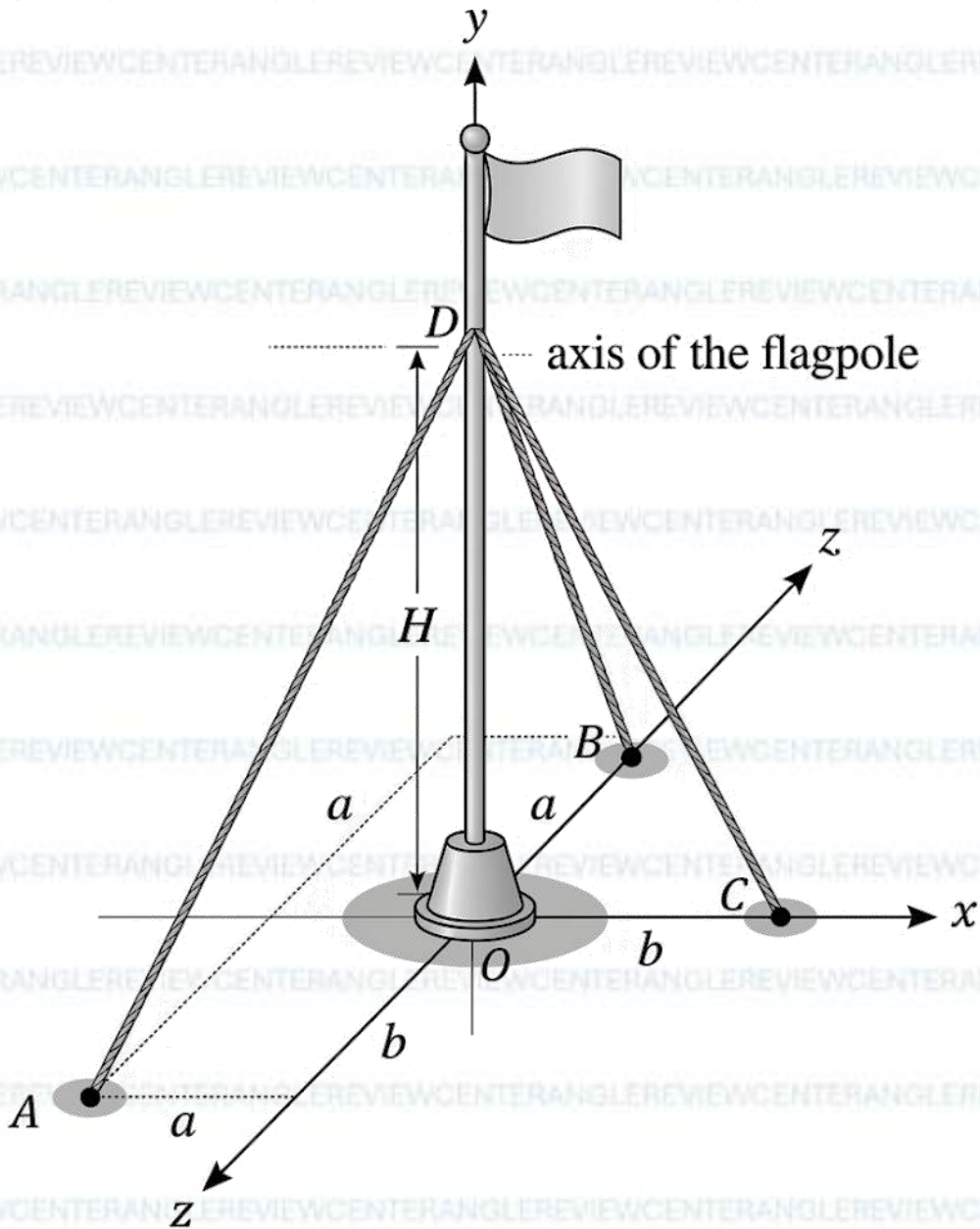
Given:

Resultant, $R = 570 \text{ kN}$

$a = 2 \text{ m}$

$b = 3 \text{ m}$

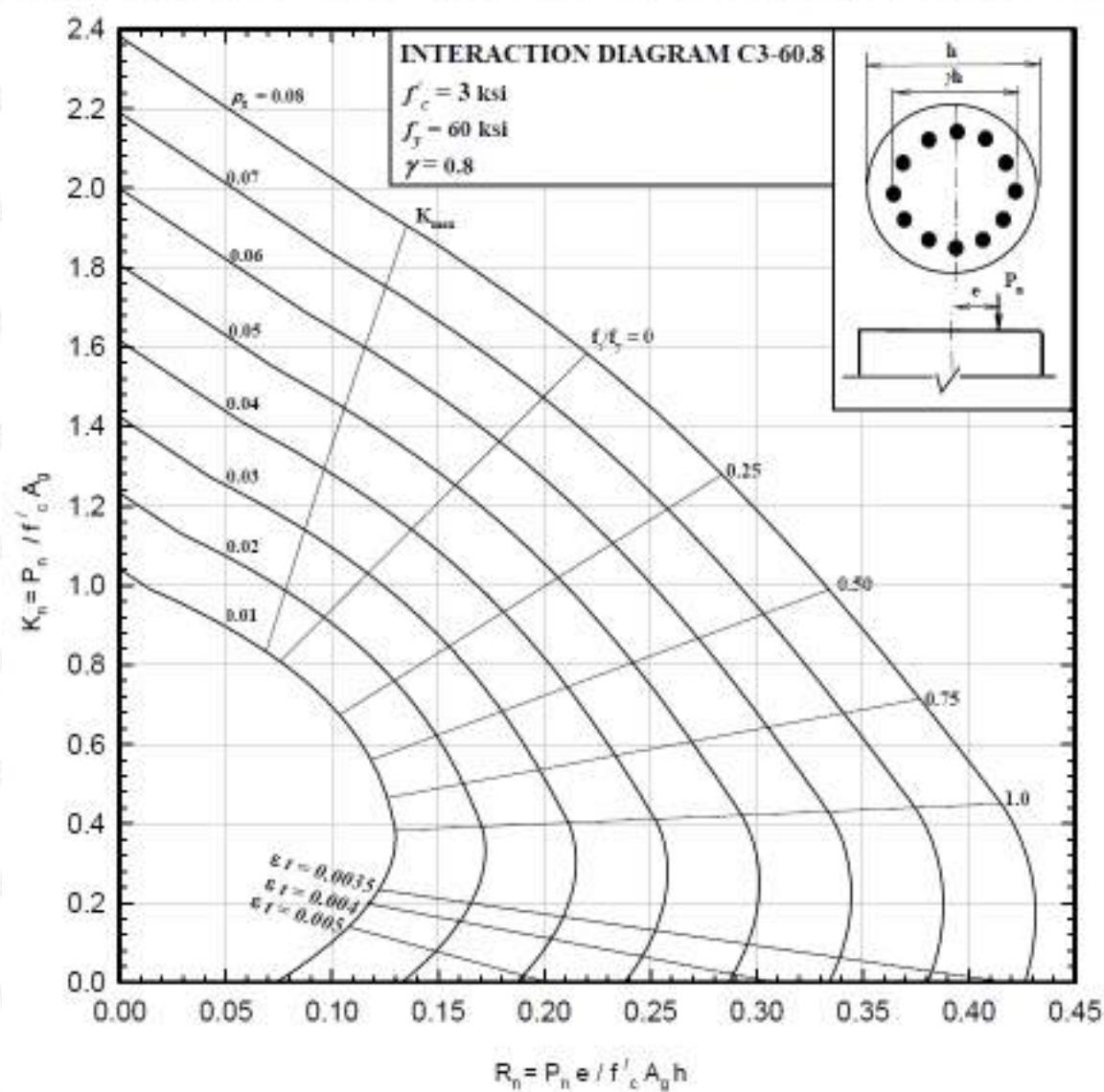
$H = 6 \text{ m}$



19. Calculate the tensile force T_1 (kN).
A. 210.0
B. 180.0
C. 245.0
D. 154.3
20. Calculate the tensile force T_2 (kN).
A. 134.2
B. 120.0
C. 201.3
D. 140.0
21. Calculate the tensile force T_3 (kN).
A. 284.4
B. 270.0
C. 301.9
D. 235.0

This review material is solely for ANGLE reviewees. Any unauthorized use, copying, distribution, sharing, or anything related to cybercrime is strictly forbidden and will be subject to legal action.

SITUATION 8 – The interaction diagram and the reinforcement layout of a circular column are shown.
 Concrete strength, $f_c' = 27.5 \text{ MPa}$
 Reinforcing steel, $f_y = 413 \text{ MPa}$
 Strength reduction factor = 0.90
 Column diameter = 600 mm
 Concrete cover to center of reinforcement = 70 mm



22. Given:

Reinforcement, $A_{st} = 12 - 25 \text{ mm diameter}$

Design axial load, $P_u = 2830 \text{ kN}$

Determine the ultimate moment capacity of the section (kN-m).

A. 840

B. 630

C. 315

D. 700
23. Given:

Design moment, $M_u = 620 \text{ kN-m}$

Design axial load, $P_u = 1850 \text{ kN-m}$

A. 22

B. 25

C. 28

D. 32
24. For the given layout, what is the REQUIRED size (mm) of reinforcing bars to provide a MINIMUM steel ratio of 3.3%?

A. 22

B. 25

C. 28

D. 32

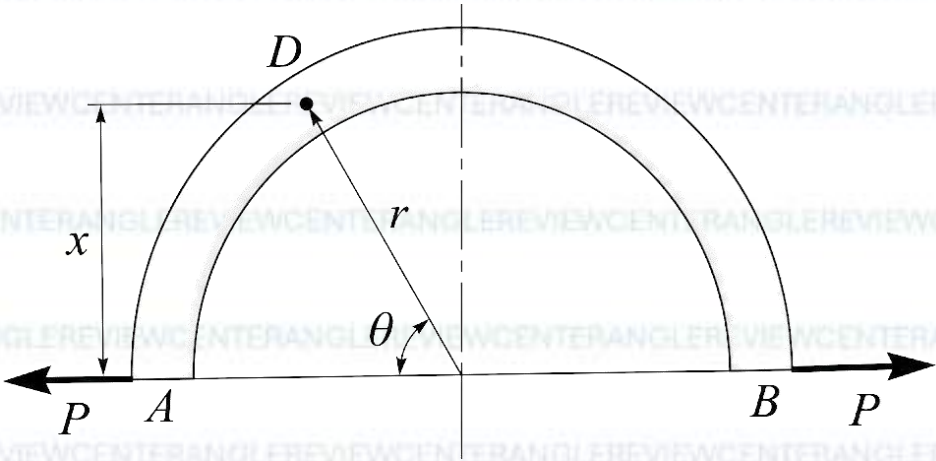
SITUATION 9 – The figure shows a semi-circular arch subjected to equal but oppositely directed forces at A and B.

Given:

$r = 1.0\text{ m}$

$x = 0.8\text{ m}$

$P = 2.8\text{ kN}$



25. Find the shear force (kN) at D.

A. 1.68

B. 2.80

C. 2.24

D. 1.12
26. Find the moment (kN-m) acting at D.

A. 1.68

B. 2.80

C. 2.24

D. 1.12
27. Find the axial force (kN) at D.

A. 1.68

B. 2.80

C. 2.24

D. 1.12

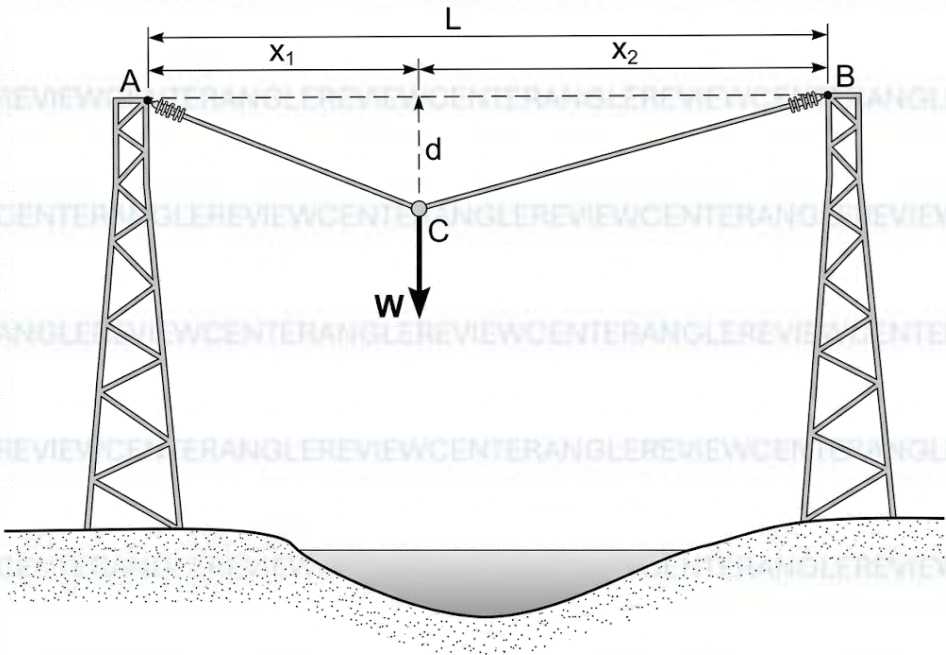
SITUATION 10 – The towers at the opposite side of the river supports the load as shown in the figure.

Dimensions:

Load, $W = 50\text{ kN}$

Distance between towers, $L = 20\text{ m}$

Sag, $d = 1.5\text{ m}$



This review material is solely for ANGLE reviewees. Any unauthorized use, copying, distribution, sharing, or anything related to cybercrime is strictly forbidden and will be subject to legal action.

28. At what distance (m) from A should the load be positioned such that the tensions in AC and AB are equal?
 A. 5
 B. 10
 C. 15
 D. 20
29. Given: $x_1 = 5$ m
 What is the tension (kN) in segment BC?
 A. 130.50
 B. 125.00
 C. 125.62
 D. 376.87
30. Given: $x_1 = 5$ m
 What is the total length (m) of the cable?
 A. 20.22
 B. 21.50
 C. 21.22
 D. 20.30

SITUATION 11 – The three-hinged arch shown in the figure is loaded with vertical loads only.

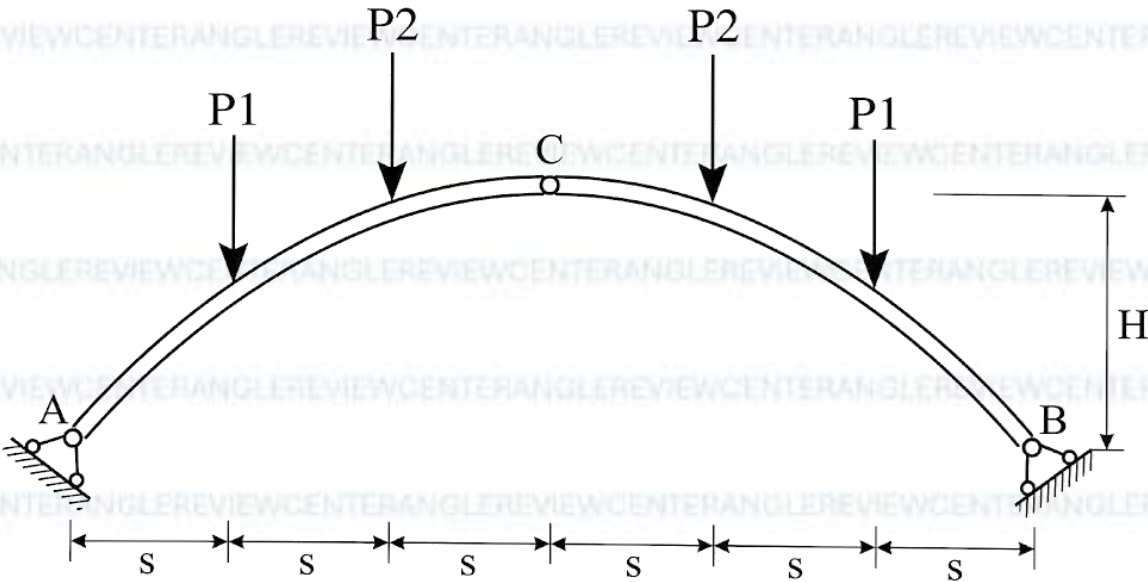
Dimensions:

$H = 12$ m

$S = 4$ m

$P_1 = 90$ kN

$P_2 = 240$ kN



31. Determine the HORIZONTAL reaction (kN) at A.
 A. 0
 B. 330
 C. 190
 D. 95
32. Determine the VERTICAL reaction (kN) at B.
 A. 330
 B. 165
 C. 0
 D. 240
33. Determine the VERTICAL reaction (kN) at C.
 A. 0
 B. 660
 C. 330
 D. 190

This review material is solely for ANGLE reviewees. Any unauthorized use, copying, distribution, sharing, or anything related to cybercrime is strictly forbidden and will be subject to legal action.

SITUATION 12 – A cylinder weighing 3.2 kN with a diameter of 2 m has a height of 3.0 m and is filled with water to a depth of 2.4 m. The coefficient 0.20. Find the MAXIMUM value of the force F (kN) located 0.6 m below the rim that will satisfy the following conditions.

34. Translation of the cylinder is prevented.

A. 25.7

B. 32.2

C. 77.1

D. 15.4
35. Tipping of cylinder is prevented.

A. 25.7

B. 32.2

C. 77.1

D. 15.4
36. Cylinder remains in equilibrium.

A. 25.7

B. 32.2

C. 77.1

D. 15.4

SITUATION 13 – A reinforced concrete cantilever beam has the following properties and is subjected to the following loads:

Dimensions, $b \times h = 250 \text{ mm} \times 400 \text{ mm}$
Length of cantilever, $L = 3 \text{ m}$
Dead load = 12 kN/m
Live load at the free end = 20 kN
Concrete strength, $f_c' = 28 \text{ MPa}$
Longitudinal steel strength, $f_y = 413 \text{ MPa}$
Lateral tie strength, $f_{yv} = 275 \text{ MPa}$
Clear concrete cover to lateral ties = 40 mm
Concrete cover to centroid of longitudinal reinforcement = 70 mm
Load factor for strength design, $U = 1.2 \text{ DL} + 1.6 \text{ LL}$
Strength reduction factors:
 Moment = 0.90
 Shear = 0.75
Diameter of lateral ties = 10 mm

37. Determine the MAXIMUM shear stress (MPa) in the cantilever beam.

A. 1.22

B. 1.00

C. 1.36

D. 2.25
38. The tension reinforcement is to be placed on one layer only. Which of the following is the MOST suitable?

A. 8 – 16 mm

B. 5 – 20 mm

C. 4 – 25 mm

D. 3 – 28 mm
39. The beam is reinforced with 2 legs of 10 mm diameter ties spaced at 200 mm on centers. Determine its nominal shear (kN) capacity.

A. 181.24

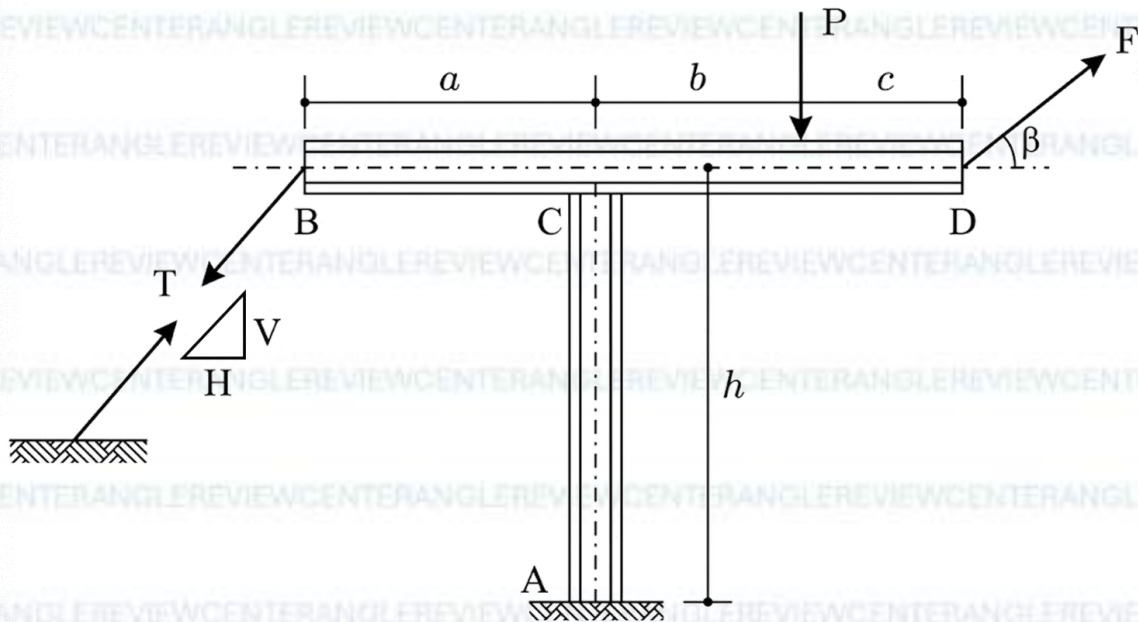
B. 109.85

C. 145.48

D. 109.11

SITUATION 14 – The beam is subjected to the following loads:

$H = 3$ $a = 0.6 \text{ m}$
 $V = 4$ $b = 1.0 \text{ m}$
 $h = 4 \text{ m}$ $c = 0.4 \text{ m}$
 $P = 60 \text{ kN}$
 $T = 150 \text{ kN}$
ANGLE, (beta) = 45 degrees
Cross-sectional area of column AC = 1990 mm²
Slenderness ratio of column AC = 138
Steel yield strength, $F_y = 415 \text{ MPa}$
Modulus of elasticity, $E = 200,000 \text{ MPa}$



Allowable stress for long column, $F_a = \frac{12 \cdot (\pi)^2 \cdot E}{23 \cdot (kL/r)^2}$
 Allowable stress for short column, $F_a = \left[1 - \frac{1}{2} \cdot \left(\frac{kL/r}{C_c} \right)^2 \right] \cdot F_y / FS$
 Factor of safety, $FS = \frac{5}{3} + \frac{3}{8} \cdot \left(\frac{kL/r}{C_c} \right) - \frac{1}{8} \cdot \left(\frac{kL/r}{C_c} \right)^3$
 CRITICAL slenderness ratio, $C_c = \sqrt{\frac{2 \cdot \pi^2 \cdot E}{F_y}}$

40. Determine the MAXIMUM capacity (kN) of column AC.
 A. 206.27
 B. 825.85
 C. 215.54
 D. 107.62
41. Determine the value of F (kN) when the capacity of column AC is MAXIMUM.
 A. 72.38
 B. 59.93
 C. 17.51
 D. 102.36
42. Determine the value of F (kN) if the moment at A is equal to zero.
 A. 152.39
 B. 12.12
 C. 127.28
 D. 230.16

SITUATION 15 – A column is supported by a square footing on 5 piles as shown.

Dimensions:

Column, $c = 600$ mm

Footing, $B = 3.6$ m

$a = 0.6$ m

$b = 2.4$ m

Effective depth, $d = 650$ mm

Diameter of pile, $D = 350$ mm

Ultimate pile capacity = 280 kN

Column axial loads:

$D = 420$ kN

$L = 360$ kN

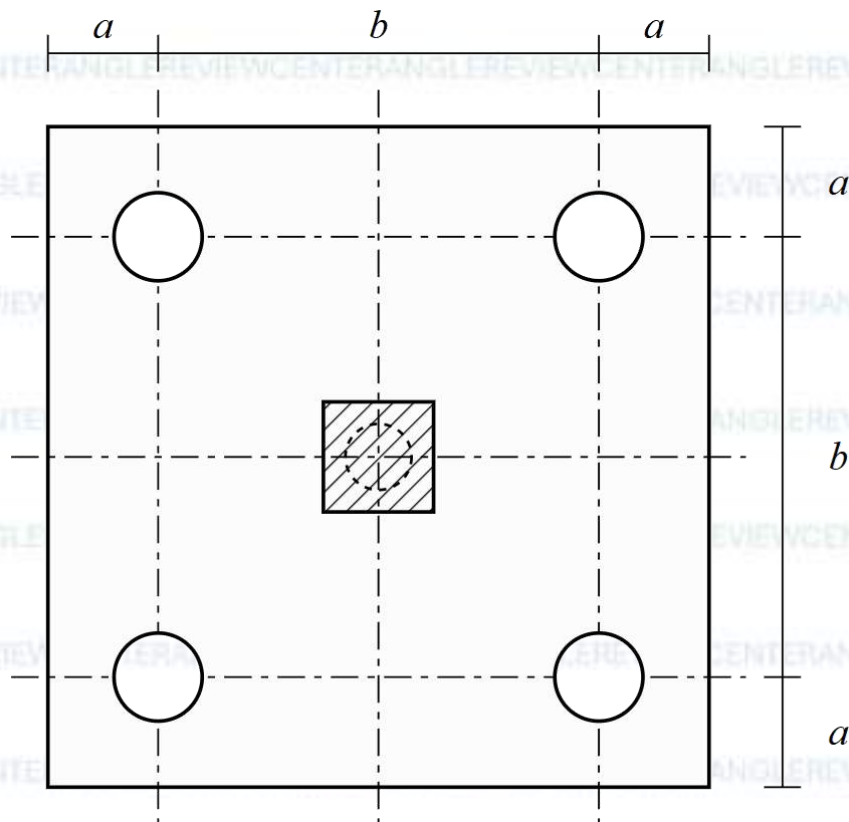
$E = 210$ kN

Column moment due to earthquake = 160 kN-m

Factored load combination, $U = 1.32 D + 1.1 L + 1.1 E$

Reduction factor for shear = 0.75

Reduction factor for moment = 0.90



43. Calculate the CRITICAL beam shear stress at ultimate loads.

A. 0.269	B. 0.311
C. 0.233	D. 0.467
44. Calculate the CRITICAL ultimate punching shear stress.

A. 0.485	B. 0.388
C. 0.291	D. 0.426
45. Calculate the punching shear of the MOST STRESSED pile.

A. 0.178	B. 0.089
C. 0.183	D. 0.176

SITUATION 16 – A steel bar carries an ore bucket weighing W hanging at its lower end.

Given:

Bar diameter = 12 mm

Length = 18 m

Steel modulus of elasticity = 200 GPa

Neglect weight of the bar.

46. Calculate the MAXIMUM weight W (kN) that can be safely carried so as not to exceed the allowable steel tensile stress of 138 MPa.

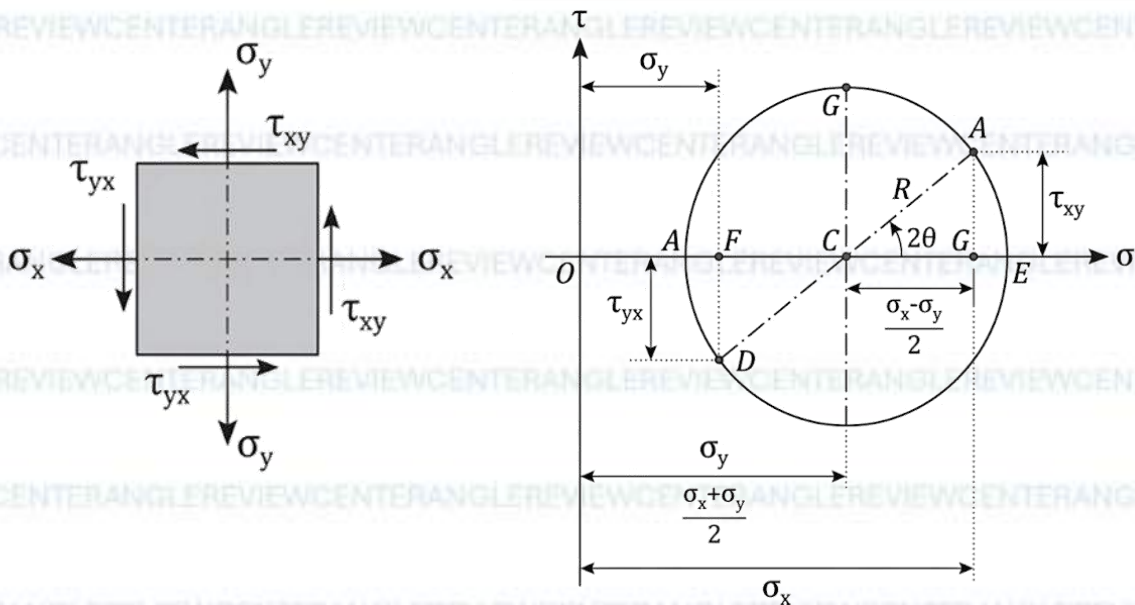
A. 15.4	B. 62.5
C. 1.22	D. 15.6
47. Calculate the MAXIMUM weight W (kN) that can be safely applied so as not to exceed the allowable elongation of 10 mm.

A. 12.6	B. 50.3
C. 15.9	D.
48. Calculate the total strain if the applied is $W = 20$ kN.

A.	B. 0.00022
C.	D. 0.00088

This review material is solely for ANGLE reviewees. Any unauthorized use, copying, distribution, sharing, or anything related to cybercrime is strictly forbidden and will be subject to legal action.

SITUATION 17 – In the Mohr’s circle for generate state of plane stress, identify the stresses described:



49. It refers to the AVERAGE of the normal stress.

A. OC

B. FC

C. AC

D. OE
50. It refers to the shear stress acting on the y-face of the element.

A. AG

B. FD

C. OF

D. CD
51. It refers to the POSITIVE tensile normal stress acting on the x-face of the element.

A. OF

B. OE

C. OG

D. OC

SITUATION 18 – Solve the following problems.

52. Determine the total length (m) of the bar.

Given:

L = 2 m

Bar diameter, db = 20 mm

Concrete over, a = 65 mm

Length of extension bar, b = 120 mm

Radius of bend, r = 4 * db

Concrete strength, fc’ = 20.7 MPa

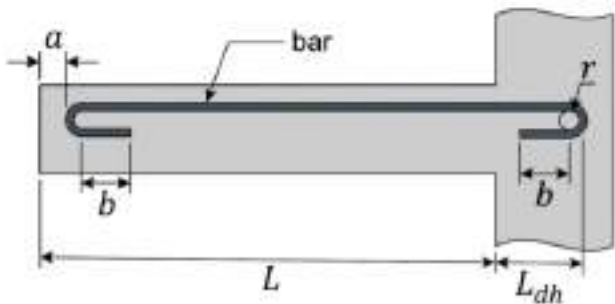
Steel yield strength, fy = 414 MPa

- 425.4.3.1 Development length “ldh” for deformed bars in tension terminating in a standard hook shall be grater of (a) through (c). Use psi = 1.0

a. (fy * psi * db) / (4.17 * sqrt(fc’))

b. 8 * db

c. 150 mm



- A. 2.53

B. 2.98

C. 3.03

D. 3.36

This review material is solely for ANGLE reviewees. Any unauthorized use, copying, distribution, sharing, or anything related to cybercrime is strictly forbidden and will be subject to legal action.

53. Given:

Thickness of one-way slab, $h = 100 \text{ mm}$

Diameter of reinforcement, $d_b = 12 \text{ mm}$

Spacing of reinforcement on centers, S = 200 mm

Concrete strength, $f_c' = 20.7$ MPa

Steel yield strength, $f_y = 275 \text{ MPa}$

Concrete cover = 20 mm

What is the DESIGN moment (kN-m) of the slab?

A. 10.82
C. 11.75

B. 10.58
D. 9.74

54. A square bar is perfectly fitted between rigid walls. Given the following data:

Length of bar = 900 mm

Bar cross-section = 16 mm x 16 mm

Coefficient of thermal expansion of the bar = $11.7 (10)^{-6}$

Modulus of elasticity, $E = 200 \text{ GPa}$

Initial temperature = 15 degrees Celsius

What force is developed in the bar when the temperature changes to 35 degrees Celsius?

A. 9.0
B. 10.2
C. 12.0
D. 13.5

SITUATION 19 – A man weighing 1.2 kN walks through a log of wood 250 mm in diameter and simply supported at a length of 3 m.

55. What is the MAXIMUM shear (kN)?

A. 0.6
B. 1.2
C. 0.8
D. 1.8

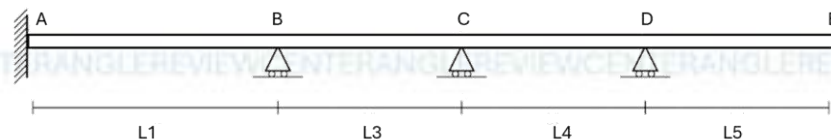
56. What is the MAXIMUM moment (kN-m)?

A. 1.5
B. 1.2
C. 0.8
D. 0.9

57. If the log is replaced with a 300 mm wide by 100 mm thick section of wood, compute the MAXIMUM weight (kN) of the man if the allowable flexural stress, $F_b = 4.2 \text{ MPa}$?

A. 4.1
C. 2.4

SITUATION 20 – The end moments of the continuous beam shown are computed using the moment distribution method. The rigidity EI is constant throughout. Span lengths:

$$L1 = 0.75 L$$
$$L2 = L$$
$$L3 = 1.2 L$$
$$L_4 = 0.25 L$$


58. Determine the distribution factor at the end A of AB.

A. 0
C. 1.0
B. 0.57
D. 0.43

59. Determine the distribution factor at the end E of DE.

A. 0
B. 0.57
C. 1.0
D. 0.43

60. Determine the distribution factor at end B for beam AB.

A. 0
B. 0.57
C. 1.0
D. 0.43

This review material is solely for ANGLE reviewees. Any unauthorized use, copying, distribution, sharing, or anything related to cybercrime is strictly forbidden and will be subject to legal action.

Initial diameter = 12 mm
Initial length = 50 mm
Diameter at rupture = 11 mm
Load at proportional limit = 28 kN
Elongation at proportional limit = 0.068 mm
Load at rupture = 55 kN

61. What is the ductility of the material, expressed as a percentage?
- A. 19.01
B. 13.60
C. 15.97
D. 16.30
62. Calculate the Young's Modulus of the material (GPa).
- A. 178
B. 156
C. 220
D. 182
63. Determine the ultimate tensile stress (MPa) of the material.
- A. 580
B. 670
C. 346
D. 486

64. Compute the tangential hoop stress (MPa).
A. 35
C. 45
B. 70
D. 90

65. Compute the longitudinal hoop stress (MPa).
A. 35
C. 45
B. 70
D. 90

66. Given:
Allowable tensile capacity of bolts = 52.5 kN
Compute the allowable internal pressure (MPa). Use a factor of safety of 1.50.
A. 0.74
C. 1.67
B. 1.11
D. 0.56

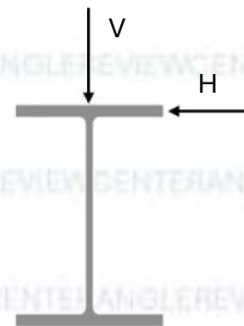
67. Solve for the resulting hoop stress (MPa).
A. 67.7
C. 33.8
B. 16.9
D. 8.5

68. Solve for the longitudinal stress (MPa) acting on the tank.
A. 67.7
C. 33.8
B. 16.9
D. 8.5

69. Given:
Allowable tensile capacity of fasteners = 52.5 kN
Factor of safety = 1.5
The cover of the tank is held in place by 24 fasteners. Solve for the MAXIMUM internal pressure (MPa) which the tank can withstand.
A. 16.7
C. 33.4
B. 25.1
D. 37.6

Reduction factor for shear = 0.75

- $$H = 8 \text{ kN}$$



- ## Web-based TQDS